Inho Yoo

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EDUCATION

Kwansei Gakuin University | Hyogo, Japan
Bachelor of Science in Physics and Astronomy | Expected March 2027
Advisor / Senior Thesis Supervisor: Professor Shuji Matsuura
Relevant Coursework: Astrophysics, Infrared Astronomy, X-ray Astronomy, Computer Science

RESEARCH INTERESTS

Planetary Surface Composition Inference • Infrared and X-ray Remote Sensing • Scientific Machine Learning • Space-Resource Exploration

RESEARCH EXPERIENCE

Undergraduate Researcher | Matsuura Laboratory, Kwansei Gakuin University

Senior Thesis: Lunar Surface Composition Inference *(Ongoing)*

- Use Kaguya MI's lunar surface visible and near-infrared reflectance data and chemical analysis results from lunar samples collected during Apollo and Chinese Chang'e-5/6 missions.
- Trained ridge regression and convolutional neural networks (CNNs) to predict FeO and TiO_2 abundances; conducted regression experiments for six major lunar oxides (SiO_2, Al_2O_3, FeO, MgO, CaO, TiO_2).
- Applied leave-one-landing-site-out (LOLO) validation by excluding data from one landing site during training and then predicting the composition of that site.
- Found promising results for FeO and Al_2O_3 compared with a baseline that predicts the mean abundance; did not find the same level of improvement for MgO, and performance varied by landing site.
- Investigated laboratory spectra and chemical analysis data from terrestrial samples; reliable matching of spectra and composition values from the same samples still needs validation.
- Analyzed Chandrayaan-2 CLASS and XSM observations and worked on linking them with Kaguya MI data; have not yet verified whether adding X-ray data improves composition prediction performance.
- Identified spatial correspondence between sample collection locations and orbital observation areas as a key limitation when using sample chemistry as ground-truth values.

SPHEREx Target List of Ice Sources (SPLICES): Spatial Distribution Analysis

- Analyzed the SPLICES FITS catalog provided by Caltech/IPAC IRSA using Python; used Astropy memory-mapped access to sample 10,000 sources from a 3.1 GB catalog containing approximately 9 million sources.
- Checked candidate coordinates and examined their spatial distribution in right ascension and declination, including the Galactic-plane pattern in equatorial coordinates.

SELECTED RESEARCH PROPOSAL

Multimodal Compositional Inference for Space-Resource Exploration *(Proposed; not validated)*

- Proposed combining infrared and X-ray data with machine learning for planetary surface composition inference and future exploration prioritization, with possible cross-body comparisons from the Moon to Bennu and Ryugu.

TECHNICAL SKILLS

Programming: Python | **Scientific Computing:** Astropy, NumPy, Pandas, Matplotlib, PDS4 Tools
Methods: FITS catalog analysis, spectral-data analysis, ridge regression, CNNs, LOLO validation
Tools: LaTeX, Visual Studio Code | **Languages:** Korean, Japanese, English